E08 Multiple Reactions in an Isothermal CSTR, 2 ways

KIA^c

The following reactions are taking place in liquid phase in a CSTR:

3A --> B + C $\quad -r_{1A} = k_{1A}C_A, k_{1A} = 7.0\ \text{min}^{-1}$

2C + A --> 3D $\quad -r_{2D} = k_{2D}C_C^2C_A, k_{2D} = 3.0\ \frac{\text{L}^2}{\text{mol}^2\text{min}}$

4D + 3C --> 3E $\quad r_{3E} = k_{3E}C_DC_C, k_{3E} = 2.0\ \frac{\text{L}}{\text{mol min}}$

The entering volumetric flowrate is 100 L/min and the entering concentration of A is 3 mol/L. The CSTR operates at C_A = 0.10 M, C_B = 0.93 M, C_C = 0.51 M, C_D = 0.049 M.

1) Determine the CSTR volume.

We get the volume from the mole balances

Here we have 5 species, we can use the mole balance for any of them

At first glance, species E might be a good choice, since It is only involved in one reaction

However, we are not given its concentration so its mole balance doesn't give us enough information

Looking again, species B only is involved in one reaction

Writing the net rate of species B and solving:

$$r_{B,net} = r_{1B}.$$

$$\frac{r_{1B}}{1} = \frac{r_{1A}}{-3}, \quad \therefore \quad r_{1B} = -\frac{1}{3} r_{1A} = \frac{1}{3} k_{1A} C_A,$$

mole balance on B:

$$V = \frac{F_{B0} - F_B}{-r_{B,net}}. \quad F_{B0} = 0, \; F_B = \underset{0.93 \frac{mol}{L}}{C_B} \; \underset{100 \frac{L}{min}}{v_0},$$

$$-r_{B,net} = -\frac{7}{3\,min}\left(\underset{C_A}{0.10 \frac{mol}{L}}\right)$$

$$\therefore \quad V = \frac{\left(100 \frac{L}{min}\right)\left(0.93 \frac{mol}{L}\right)}{\left(\frac{7}{3\,min}\right)\left(0.10 \frac{mol}{L}\right)} = 399\,L$$

2) Let's pretend we don't know any of the exit concentrations, but we know the volume is 400 L. Determine the exit concentrations.

Now, the number of unknowns = 5

We hence need 5 equations. The equations that we will use are the

mole balances

These take the form $V = \frac{F_{j0} - F_j}{-r_{j,net}}$

Alternatively, we can write them

$$0 = F_{j0} - F_j + V r_{j\,net}$$

Setting them equal to zero sets us up well to use a non-linear equation solver such as fsolve, which finds the "roots" of an equation

To do this, first let's write the

the stoichiometry matrix & the rate matrix

$$r = \begin{bmatrix} k_{1A} C_A \\ k_{2D} C_C^2 \, C_A \\ k_{3E} C_D \, C_E \end{bmatrix}$$

reactions:

$$A \longrightarrow \tfrac{1}{3} B + \tfrac{1}{3} C$$

$$\tfrac{2}{3} C + \tfrac{1}{3} A \longrightarrow D$$

$$\tfrac{4}{3} D + C \longrightarrow E$$

stoichiometry matrix:

$$r = \begin{bmatrix} -1 & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ -\frac{1}{3} & 0 & -\frac{2}{3} & 1 & 0 \\ 0 & 0 & -1 & -\frac{4}{3} & 1 \end{bmatrix}$$